

Md. Akash Miya

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PROFESSIONAL PROFILE

B.Sc. (Engineering) graduate in Information and Communication Engineering from Pabna University of Science and Technology with a CGPA of 3.47/4.00. Research experience includes Artificial Intelligence, Deep Learning, Computer Vision, Biomedical Imaging, Signal Processing, and Explainable AI. Published research works in IEEE and Springer platforms, and experienced in ICT instruction.

EDUCATION

Bachelor of Science in Engineering (B.Sc. Engg.) in Information and Communication Engineering | *Pabna University of Science and Technology (PUST)* 2026

- Undergraduate Thesis: "Annotation Guided and Uncertainty Aware Explainable Multi Task Learning for Active Pulmonary Tuberculosis Detection and Lesion Localisation."

Higher Secondary Certificate (HSC) – Science | *M.E.H. Arif College, Gazipur* 2019

• GPA: 4.42/5.00
Secondary School Certificate (SSC) – Science | *Mouchak Ideal Public School and College, Gazipur* 2017

- GPA: 5.00/5.00

ACADEMIC DISTINCTION

Academic Excellence Award – Secondary School Certificate Examination (SSC), 2017 — Received recognition for outstanding academic performance in SSC examination by LIZ Fashion Industry Limited, Gazipur, Dhaka, Bangladesh.

PUBLICATIONS & RESEARCH DEVELOPMENT

- Remote Health Monitoring via PPG Signal Processing and Machine Learning Using Mobile Devices.** 2025 28th International Conference on Computer and Information Technology (ICCIT), IEEE, 2026. **First Author.**
- Machine Learning-Based Hepatitis C Detection Using Ensemble Boosting.** *Lecture Notes in Networks and Systems (LNNS)*, Springer Nature, 2026.
- Machine Learning-Based Prediction of Nightmares in Children: A Pediatric Psychological Integration.** 2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS), IEEE, 2026.
- GR-ACE Net: A Hybrid Graph-Attentional Framework with Global Relational Reasoning for Deepfake Forensics.** 2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII), IEEE.
- A Comparative Study of CNN, Transformer, and Recurrent Models for Deepfake Detection on Degraded Images.** 2026 International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII), IEEE.
- Comparative Evaluation of CNN Backbones with Explainable AI for Multi-Class Chest X-Ray Classification.** 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), IEEE.
- A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images.** 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN), IEEE.

Awards & Achievements

- Academic Excellence Recognition:** Received recognition for outstanding academic performance in SSC examination by LIZ Fashion Industry Limited, Gazipur, Dhaka, Bangladesh.
- 2nd Runner-Up:** Programming Contest 2025, Department of Information and Communication Engineering, PUST.
- Research Paper Presentation Recognition, IEEE ICCIT 2025 :** Presented "Remote Health Monitoring via PPG Signal Processing and Machine Learning Using Mobile Devices."
- Research Paper Presentation Recognition, IEEE COMPAS 2025 :** Presented "Machine Learning-Based Prediction of Nightmares in Children: A Pediatric Psychological Integration."
- Research Paper Presentation Recognition, BIM 2025 (Springer/Taylor & Francis supported) – :** Presented "Machine Learning-Based Hepatitis C Detection Using Ensemble Boosting."
- Research Paper Presentation Recognition, IEEE QPAIN 2026 – :** Presented research on "A Hybrid YOLO-EfficientNet Framework for Fine-Grained Vegetable Recognition in Dense Multi-Object Images."

PROFESSIONAL EXPERIENCE

ICT Instructor | *Srizon Coaching Center, Pabna* 2023 – Present

- Teaching ICT courses and preparing lesson materials for students.
- Conducting classes and assessments and providing academic guidance.
- Helping students develop strong ICT knowledge and problem-solving skills.

TECHNICAL & RESEARCH SKILLS

Programming: Python, C, C++, Java, JavaScript, PHP, SQL, MATLAB
AI & Computer Vision: PyTorch, TensorFlow, Keras, OpenCV, NumPy, Pandas, Matplotlib
Research & Data: Field image acquisition, bounding-box annotation, dataset curation and splitting, model training and evaluation, literature review, scholarly writing
Systems & Tools: Jupyter Notebook, Kaggle, Visual Studio Code, SQL Server Management Studio, Proteus, mikroC, Cisco Packet Tracer

SELECTED ACADEMIC COURSEWORK

Artificial Intelligence and Robotics • Neural Networks • Natural Language Processing • Digital Image and Speech Processing • Digital Signal Processing • DBMS • Web Programming • Software Analysis and System Testing • Wireless Communications • Computer Architecture and Microcontroller Design

PROJECTS, CERTIFICATIONS & LEADERSHIP

Projects & Certification: SQL (Basic) - HackerRank; Universal Control Board System (Microcontroller and hardware interfacing).
Join General Secretary | *PUST Career and Entrepreneurship Club* 2025 - 2026
Assistant Research and Course Development Secretary | *Solver Green, PUST* 2024 - 2025

REFERENCES

Dr. Md. Anwar Hossain Professor, Department of Information and Communication Engineering, PUST <i>Thesis Supervisor</i> +880 1717-330923 manwar.ice@pust.ac.bd	Dr. Md. Sarwar Hossain Professor, Department of Information and Communication Engineering, PUST <i>Academic Referee</i> +880 1722 047833 sarwar.ice@pust.ac.bd
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